

Noise-Robust Motion Compensation for Aerial Maneuvering Target ISAR Imaging by Parametric Minimum Entropy Optimization

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Abstract—When a target is involved in maneuvering motion, the nonuniform 3-D rotation motion will cause a continuous change of image projection plane (IPP), which would induce 2-D spatial-variant phase errors. In this case, the inverse synthetic aperture (ISAR) image would be seriously blurred when using the traditional compensation methods. On the other hand, strong noise has been always challenging the conventional methods in motion parameters estimation and phase error compensation. In this paper, we propose a noise-robust compensation method to compensate the 2-D spatial-variant phase errors of the maneuvering target via using tracking information and parametric minimum entropy optimization. First, the maneuvering signal model is developed based on a 2-D spatial-variant model and a 3-D rotation motion model. Based on the developed signal model, a parametric entropy minimum optimization is established to estimate the rotation motion parameters. A gradient-based solver of this optimization is then adopted to iteratively find the global optimum. Meanwhile, in order to increase the robustness of this optimization under low SNR, an extended Kalman filter is adopted here for coarse motion estimation via using tracking information. By treating these estimated motion parameters as initial values, we can effectively prevent this optimization from trapping into a local optimum. Finally, the 2-D spatial-variant phase error can be iteratively compensated, and a well-focused ISAR image can be obtained. The proposed method has three main contributions: 1) it is applicable in the case of changing IPP; 2) it gives the exact expression of chip parameters; and 3) it can efficiently compensate the 2-D spatial-variant phase errors under low SNR. Experiments based on the simulated data and the real measured data prove the effectiveness and robustness of the proposed method.

Index Terms—2-D spatial-variant phase errors, inverse synthetic aperture radar (ISAR), minimum entropy, motion compensation, noise robust.

Manuscript received April 26, 2018; revised August 12, 2018; accepted November 21, 2018. Date of publication January 25, 2019; date of current version June 24, 2019. This work was supported in part by the 111 Project under Grant B18039 and in part by the National Science Foundation of China under Grant U1833203, Grant 61771362, and Grant 61671354. (*Corresponding author: Lan Du.*)

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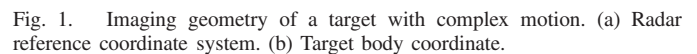
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Digital Object Identifier 10.1109/TGRS.2018.2890098

I. INTRODUCTION

INVERSE synthetic aperture radar (ISAR) has found a wide application in both civilian and military fields due to its capability of providing 2-D high-resolution image of moving targets. In general, the high range resolution depends on transmitting wideband waveforms, whereas the high azimuth resolution depends on both the coherent processing interval (CPI) and the relative motion between the radar and the observed targets. Conventionally, the primary procedures for ISAR imaging are the range migration compensation and the translational-induced phase error correction. Thereafter, the traditional range-Doppler (RD) method is utilized to reconstruct the images for targets [1]. However, the RD method is only suitable for targets with smooth motions and cannot handle extreme situations, for example, ISAR imaging for nonuniformly rotating targets. In the practical ISAR imaging, the maneuvering targets (such as highly maneuvering airplanes and fluctuating ships with oceanic waves) are generally engaged in complicated motion, such as 3-D angular motion (roll, pitch, and yaw). In this circumstance, maneuvering property challenges the performance of the conventional RD algorithm in several aspects. One of the challenges is that nonuniform 3-D rotational motion will cause a continuous change in the direction and the modulus value of the effective rotation vector, which would lead to the continuous change of image projection plane. As a result, the phase errors exhibit the 2-D spatial-variant characteristic [2]. A high-resolution ISAR image is very sensitive to range and azimuth spatial-variant phase errors, which are not only connected with the fast and slow time but also correlated with the position of scattering centers in the ISAR imagery. If the 2-D spatial-variant phase errors are not corrected properly, the final image may get blurred when the RD algorithm is applied. Another challenge is the strong noise. In radar remote sensing, due to the signal decay from long range and absorption of transmitting medium, strong noise usually presents. The SNR problem is among the most significant challenges that ISAR imaging systems frequency face. In the presence of strong noise, the precise estimation of rotational motion parameters during motion compensation always encounters some difficulties. As a result, the performance of the conventional imaging algorithm would degrade seriously [3]–[5].

In this paper, a novel noise-robust algorithm of ISAR rotational motion compensation for the maneuvering target is proposed. It should be pointed out that the integrity of the target motion is taken into account, and only two parameters are needed to be estimated during the whole imaging process, which would increase the accuracy of parameter estimation to a great extent. After the range compression and translational motion compensation, the Doppler modulation of radar echoes is mainly caused by the target's complex rotational motion. We formulate the Doppler modulation of the target as range and azimuth dependence, which exhibits the 2-D spatial-variant characteristic. Then, a general and exact parametrical signal model of phase errors is established in our proposed algorithm. This signal model contains two components: range spatial-variant phase errors and azimuth spatial-variant phase errors. Based on the signal model, we build the parametric entropy model for ISAR image, which turns the problem of phase errors compensation into a minimization optimization problem. To solve the minimization optimization, a gradient-based optimization algorithm is used to search the global optimum solution for the 2-D spatial-variant phase errors. To avoid trapping into local optimum under low SNR, we also propose a method to coarsely estimate the rotational motion parameters based on an extended Kalman filter (EKF) via



This paper is organized as follows. In Section II, we establish the ISAR imaging geometry and the signal model. In Section III, the algorithm based on entropy minimum optimization is developed, and the gradient-based solver of this optimization problem is introduced. In Section IV, some imaging results based on the simulated data and the measured data are given, and the performance of the ISAR imaging method is analyzed. Some conclusions are given in Section V.

In this section, we present necessary introduction and assumptions on ISAR imaging geometry and target 3-D motion to facilitate the following discussions. Fig. 1(a) shows the ISAR imaging geometry. The target is placed in a Cartesian coordinate (X, Y, Z) , called body coordinates of the target, with the origin O as the rotating center and the reference imaging center. The (X_s, Y_s, Z_s) is the radar reference coordinate system, which acts as the absolute reference system with O_s being the origin. R_a denotes the distance between the radar and the target. The target is observed by the radar at an elevation angle β and an azimuth angle α at the starting time of the considered imaging interval. Fig. 1(b) describes the target

body coordinate. The target is placed in the XOY plane; the red solid line represents the radar line-of-sight (LOS), which is determined by the azimuth and elevation angles η_l and φ_l . The LOS is unlikely to change in a short imaging interval. Without loss of generality, the following discussions depend on the following assumptions.

- 1) The three components of target's 3-D rotational motion, namely roll, pitch, and yaw, are all modeled as the second-order series with a constant rotational velocity (RV) and a constant rotational acceleration (RA).
- 2) In the target body coordinates, the airplane can be considered as a flat target.

In Section II-A, we would introduce the signal model of nonuniformly rotating targets based on the 3-D rotational motion model. And the characteristic of 2-D spatial-variant phase errors would also be discussed.

A. Signal Model of Nonuniformly Rotating Targets

In ISAR imaging, the linear-frequency-modulated signals are adopted in the imaging radar, i.e.,

$$s(\hat{t}; t_m) = \text{rect}\left(\frac{\hat{t}}{T_P}\right) \cdot \exp\left[j2\pi\left(f_c t + \frac{1}{2}\gamma t^2\right)\right] \quad (1)$$

where $\text{rect}(u) = \begin{cases} 1, & |u| \leq \frac{1}{2} \\ 0, & |u| > \frac{1}{2} \end{cases}$ and T_P , f_c , and γ denote the pulsewidth, carrier frequency, and frequency modulation rate, respectively. $t = \hat{t} + t_m$ is the full time, where $\hat{t}$ is the fast time and t_m is the slow time. As shown in Fig. 1(b), point P is a random scatterer on the airplane, whose coordinate can be denoted as (x_p, y_p) in the target body coordinate according to assumption 2). Suppose that the distance between the scatterer P and imaging radar at the time t_m is $R_p(t_m)$, then the radar echo of this scatterer can be written as

$$\begin{aligned} s(\hat{t}; t_m) &= \sigma_p \text{rect}\left(\frac{\hat{t} - \frac{2R_p(t_m)}{c}}{T_P}\right) \\ &\cdot \exp\left[j2\pi\left(f_c\left(t - \frac{2R_p(t_m)}{c}\right) + \frac{1}{2}\gamma\left(\hat{t} - \frac{2R_p(t_m)}{c}\right)^2\right)\right] \end{aligned} \quad (2)$$

where c and σ_p represent the speed of light and the reflection coefficient, respectively. After the dechirping technique [20] with a replica of (1) and range compression, the radar echoes become

$$\begin{aligned} s_p(\hat{t}; t_m) &= \sigma_p \cdot \text{sinc}\left[B \cdot \left(\hat{t} - \frac{2R_p(t_m)}{c}\right)\right] \\ &\cdot \exp\left[-j2\pi f_c \frac{2R_p(t_m)}{c}\right] \end{aligned} \quad (3)$$

where $B = \gamma T_P$ is the bandwidth, $\lambda = c/f_c$ is the wavelength, and $\text{sinc}(x) = ((\sin(\pi x))/\pi x)$. By applying the Fourier transform with respect to $\hat{t}$ and the matched filtering to the signal, we have

$$s_p(f_r; t_m) = \sigma_p \cdot \exp\left[-j2\pi \frac{2R_p(t_m)}{c}(f_r + f_c)\right] \quad (4)$$

In general, the instantaneous range between the radar and scatterer P can be expressed as

$$R_p(t_m) = R_a(t_m) + R_{\text{rot}}(t_m) \quad (5)$$

where $R_a(t_m)$ corresponds to the translational motion part; $R_{\text{rot}}(t_m)$ refers to the rotational motion part, which can be expressed as (6) using a rotation matrix $\mathbf{rot}(t_m)$ [21], [22]

$$R_{\text{rot}}(t_m) = [\mathbf{rot}(t_m) \cdot \mathbf{P}]^T \cdot \mathbf{i}_{\text{los}} \quad (6)$$

where $\mathbf{P} = [x_p, y_p, 0]^T$ is the position vector of point P in the target body coordinate, $\mathbf{i}_{\text{los}} = [\cos \varphi_l \cos \eta_l, \cos \varphi_l \sin \eta_l, \sin \varphi_l]^T$ is the unit vector along the radar LOS, and $\mathbf{rot}(t_m)$ can be expressed as follows [23]:

$$\mathbf{rot}(t_m) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (7)$$

where

$$\begin{cases} a_{11} = \cos \theta_p(t_m) \cos \theta_y(t_m) \\ a_{12} = -\cos \theta_p(t_m) \sin \theta_y(t_m) \\ a_{13} = \sin \theta_p(t_m) \\ a_{21} = \sin \theta_r(t_m) \sin \theta_p(t_m) \cos \theta_y(t_m) \\ \quad + \cos \theta_r(t_m) \sin \theta_y(t_m) \\ a_{22} = -\sin \theta_r(t_m) \sin \theta_p(t_m) \sin \theta_y(t_m) \\ \quad + \cos \theta_r(t_m) \cos \theta_y(t_m) \\ a_{23} = -\sin \theta_r(t_m) \cos \theta_p(t_m) \\ a_{31} = -\cos \theta_r(t_m) \sin \theta_p(t_m) \cos \theta_y(t_m) \\ \quad + \sin \theta_r(t_m) \sin \theta_y(t_m) \\ a_{32} = \cos \theta_r(t_m) \sin \theta_p(t_m) \sin \theta_y(t_m) \\ \quad + \sin \theta_r(t_m) \cos \theta_y(t_m) \\ a_{33} = \sin \theta_r(t_m) \cos \theta_p(t_m). \end{cases} \quad (8)$$

It can be seen from (6) that, for a target with rotational motion, the variation of $R_{\text{rot}}(t_m)$ is mainly induced by angular motion roll, pitch, and yaw. For maneuvering targets, the rotation rate is not constant, leading to the fact that the rotation angles of roll, pitch, and yaw are not uniformly changed. According to assumption 1) and [24], the three rotation angles can be expressed as

$$\begin{cases} \theta_r(t_m) \approx \omega_r t_m + \frac{\dot{\omega}_r}{2} t_m^2 \\ \theta_p(t_m) \approx \omega_p t_m + \frac{\dot{\omega}_p}{2} t_m^2 \\ \theta_y(t_m) \approx \omega_y t_m + \frac{\dot{\omega}_y}{2} t_m^2 \end{cases} \quad (9)$$

where ω_r , ω_p , and ω_y are the constant RV, $\dot{\omega}_r$, $\dot{\omega}_p$, and $\dot{\omega}_y$ are the RA of roll, pitch, and yaw, respectively. In general, the higher order terms can be ignored, and the approximated rotation angles denoted by (9) can satisfy most situations when maneuverability of the noncooperative target is not too severe. For much severer maneuvering targets, the CPI should be limited to satisfy the condition denoted by (9).

By substituting (9) into (8), we can get the expression of the rotation matrix as (11), as shown at the top of the next page,

$$\mathbf{rot}(t_m) \approx \begin{bmatrix} 1 - \frac{\omega_p^2 + \omega_y^2}{2} t_m^2 & -\omega_y t_m - \frac{\dot{\omega}_y}{2} t_m^2 & \omega_p \cdot t_m + \frac{\dot{\omega}_p}{2} t_m^2 \\ \omega_r \omega_p t_m^2 + \omega_y t_m + \frac{\dot{\omega}_y}{2} t_m^2 & 1 - \frac{\omega_r^2 + \omega_y^2}{2} t_m^2 & -\omega_r t_m - \frac{\dot{\omega}_r}{2} t_m^2 \\ \omega_r \omega_y t_m^2 - \omega_p t_m - \frac{\dot{\omega}_p}{2} t_m^2 & \omega_y \omega_p t_m^2 + \omega_r t_m + \frac{\dot{\omega}_r}{2} t_m^2 & 1 - \frac{\omega_p^2 + \omega_r^2}{2} t_m^2 \end{bmatrix} \quad (11)$$

where (11) is based on the two-order Taylor series expansion

$$\begin{aligned} \cos(\theta) &\approx 1 - \frac{\theta}{2} \approx 1 - \frac{\omega^2 t_m^2}{2} \\ \sin(\theta) &\approx \theta \approx \omega t_m + \frac{\dot{\omega}}{2} t_m^2 \end{aligned} \quad (10)$$

where θ represents the rotation angle and ω and $\dot{\omega}$ represent the RV and RA of roll, pitch, and yaw respectively. Substituting (11), shown at the top of this page, back into (6) and for the convenience of description, $\mathbf{i}_{\text{los}}$ is re-expressed as $\mathbf{i}_{\text{los}} = [i_1, i_2, i_3]^T$, where $i_1 = \cos \varphi_l \cos \eta_l$, $i_2 = \cos \varphi_l \sin \eta_l$, and $i_3 = \sin \varphi_l$; $R_{\text{rot}}(t_m)$ can be expressed as

$$R_{\text{rot}}(t_m) \approx K_0 + K_1 t_m + K_2 t_m^2 \quad (12)$$

where K_0 , K_1 , and K_2 are shown at the top of the next page.

Using (20), K_2 is linear with respect to K_0 and K_1 , which is given by

$$K_2 = \zeta_y K_0 + \zeta_x K_1 \quad (13)$$

where detailed forms of $[\zeta_y, \zeta_x]^T$ can be found in Appendix A. Substituting (12) and (13) back into (5), the instantaneous range between the radar and scatter P can be re-expressed as

$$\begin{aligned} R_p(t_m) &= R_a(t_m) + R_{\text{rot}}(t_m) \\ &\approx R_a(t_m) + K_0 + K_1 t_m + (\zeta_y K_0 + \zeta_x K_1) t_m^2. \end{aligned} \quad (14)$$

Finally, the echo signal of the nonuniform rotational target can be written as

$$\begin{aligned} s_p(f_r; t_m) &\approx \sigma_p \cdot \exp \left[-j 2\pi \frac{2R_a(t_m)}{c} (f_r + f_c) \right] \\ &\cdot \exp \left[-j 4\pi \frac{K_0 + K_1 t_m + (\zeta_y K_0 + \zeta_x K_1) t_m^2}{c} (f_r + f_c) \right]. \end{aligned} \quad (15)$$

B. 2-D Spatial-Variant Phase Errors

In general, (15) can be rewritten as

$$\begin{aligned} s_p(f_r; t_m) &\approx \sigma_p \cdot \exp \left[-j \frac{4\pi}{\lambda} K_1 t_m \right] \cdot \exp \left[-j \frac{4\pi}{c} K_0 f_r \right] \\ &\cdot \exp \left[-j 2\pi \frac{2R_a(t_m)}{c} f_c \right] \\ &\cdot \exp \left[-j 4\pi \frac{R_a(t_m) + K_1 t_m + (\zeta_y K_0 + \zeta_x K_1) t_m^2}{c} f_r \right] \end{aligned}$$

$$\begin{aligned} &\cdot \exp \left[-j \frac{4\pi}{\lambda} (\zeta_y K_0 + \zeta_x K_1) t_m^2 \right] \\ &\cdot \exp \left[-j \frac{4\pi}{\lambda} K_0 \right]. \end{aligned} \quad (16)$$

In (16), the phase is separated into six parts: the first is the linear phase term corresponding to the Doppler, the second term corresponds to the range compression, the third term causes translational motion, the fourth term corresponds to the range migration, the fifth term $\exp[-j(4\pi/\lambda)(\zeta_y K_0 + \zeta_x K_1)t_m^2]$ induces 2-D spatial-variant phase errors, which are caused by target's nonuniform rotational motion, and the last term $\exp[-j(4\pi/\lambda)K_0]$ is a constant term, which has no influence on ISAR imaging. Unless the third, fourth, and fifth terms are optimally compensated, the final image would be seriously blurred. Notice that, the translational part and the range migration are the same for all the scatterers of the target and can be compensated for all the scatterers at the same time. However, the 2-D spatial-variant phase errors are different from scatterer to scatterer, and hence, it should be compensated for every pixel differently. In general, the translational motion can be compensated by the conventional R-D algorithm effectively. However, when the target is involved in complex motion, the translational motion between the target and the radar usually has high-order terms. What is more, the received signal is always interfered by strong noise. As a result, the conventional methods may fail to compensate the translational motion. Some algorithms have been proposed to deal with this problem, such as [25]–[27]. These algorithms consider the translational motion error as high-order polynomial, which can be compensated ideally under low SNR. In the following contents, we assume that the translational motion has been compensated ideally by these methods, and we only focus on the 2-D spatial-variant phase errors term. After the translational motion compensation, the echo signal can be expressed as follows:

$$s_p(f_r; t_m) \approx \tilde{s}_p(f_r; t_m) \cdot \exp \left[-j \frac{4\pi}{\lambda} (\zeta_y K_0 + \zeta_x K_1) t_m^2 \right] \quad (17)$$

where $\tilde{s}_p(f_r; t_m)$ is the signal without phase errors which is expressed as follows:

$$\tilde{s}_p(f_r; t_m) = \sigma_p \cdot \exp \left[-j \frac{4\pi}{c} K_0 f_r \right] \exp \left[-j \frac{4\pi}{\lambda} K_1 t_m \right]. \quad (18)$$

Applying the inverse Fourier transform with respect to f_r and considering the inevitable noise, (17) can be expressed in a discrete form as

$$s_p(k; n) \approx \tilde{s}_p(k; n) \cdot \exp \left[-j \frac{4\pi}{\lambda} (\zeta_y K_0 + \zeta_x K_1) n^2 \right] + \xi(k; n) \quad (19)$$

$$\begin{aligned}
K_0 &= i_1 x_p + i_2 y_p \\
K_1 &= (i_2 \omega_y - i_3 \omega_p) x_p + (i_3 \omega_r - i_1 \omega_y) y_p \\
K_2 &= \left[-i_1 \frac{\omega_p^2 + \omega_y^2}{2} + i_2 \left(\frac{\dot{\omega}_y}{2} + \omega_r \omega_p \right) - i_3 \left(\frac{\dot{\omega}_p}{2} - \omega_r \omega_p \right) \right] x_p + \left[-i_1 \frac{\dot{\omega}_y}{2} - i_2 \frac{\omega_r^2 + \omega_y^2}{2} + i_3 \left(\frac{\dot{\omega}_r}{2} + \omega_y \omega_p \right) \right] y_p \quad (20)
\end{aligned}$$

where $k = 1, 2, \dots, K$, k is the range indices, K is the number of range bins, and $n = 1, 2, \dots, N$, where n is the aperture position and N is the number of azimuth aperture. $\tilde{s}_p(k; n)$ is the discrete form of $\tilde{s}_p(\hat{t}; t_m)$ and $\xi(k; n)$ denotes the complex noise. And the complex-valued 2-D image is obtained by applying inverse discrete Fourier transform (IDFT) to $s_p(k; n)$ in the cross-range direction, which is given by

$$\begin{aligned}
g(k; h) &= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot s_p(k; n) \\
&= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot \tilde{s}_p(k; n) \\
&\quad \cdot \exp \left[-j \frac{4\pi}{\lambda} (\zeta_y K_0 + \zeta_x K_1) n^2 \right] + \xi(k; h) \quad (21)
\end{aligned}$$

where $\xi(k; h)$ denotes the complex noise after IDFT. According to Appendix B, (21) can be re-expressed as

$$\begin{aligned}
g(k; h) &= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot s_p(k; n) \\
&= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot \tilde{s}_p(k; n) \\
&\quad \cdot \exp[j(C_x h + C_y k)n^2] + \xi(k; h) \quad (22)
\end{aligned}$$

where the detailed forms of C_x and C_y are given in Appendix B. $\exp[j(C_x h + C_y k)n^2]$ in (22) is the 2-D spatial-variant phase errors. Notice that, for different h and k values, i.e., different pixels in an ISAR image, the phase errors are completely different. By compensating this term, the focused ISAR image can be expressed as

$$\begin{aligned}
\tilde{g}(k; h) &= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot s_p(k; n) \\
&\quad \cdot \exp[-j(C_x h + C_y k)n^2] \quad (23)
\end{aligned}$$

where $\tilde{g}(k; h)$ is the focused image and $s_p(k; n)$ is the polluted azimuth phase history data that need to be corrected. $\exp[-j(C_x h + C_y k)n^2]$ is the phase error compensation term; C_x and C_y describe the feature of azimuth and range spatial-variant features, respectively. As we can note that if the 2-D spatial-variant phase error is completely compensated, the final image $\tilde{g}(k; h)$ can actually be expressed as follows:

$$\tilde{g}(k; h) = IFFT_{t_m} \{ FFT_{f_r} [\tilde{s}_p(f_r; t_m)] \}. \quad (24)$$

The final image is the same as that obtained by the RD algorithms. (23) gives a closed-form solution of the relationship between the polluted azimuth phase history data and the focused image, and it can be used for 2-D spatial-variant phase

errors compensation from the unfocused ISAR imagery if the parameters set (C_x, C_y) is known exactly. So, in the following, the key task is to find the optimum solution of these two parameters.

III. OPTIMAL PARAMETERS ESTIMATION

Section III-A will introduce the optimization algorithm based on the parametric minimum entropy. To increase the robustness of the algorithm under low SNR, we also propose a coarse motion estimation method to promote the convergence of the optimization.

A. Optimization Based on Parametric Minimum Entropy

Entropy is one of the most primary functions in error compensation for ISAR imaging, and it has been used as a typical indicator in ISAR imaging in many different ways [28]–[30]. The entropy of the 2-D image represents its sharpness, and generally, the “sharpest” image is corresponding to the fully focused image. Inspired by [24], the entropy is adopted in this paper to measure the focus quality of the compensated ISAR imagery. According to (23), a focused ISAR image can be achieved by compensating the polluted azimuth phase history data with the phase error compensation term $\exp[-j(C_x h + C_y k)n^2]$, and the complex image after phase error compensation can be rewritten as

$$\begin{aligned}
\tilde{g}(k, h; C_x, C_y) &= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot s_p(k; n) \\
&\quad \cdot \exp[-j(C_x h + C_y k)n^2]. \quad (25)
\end{aligned}$$

Therefore, the entropy of an image is defined as a function of (C_x, C_y) , which is given by

$$\begin{aligned}
E_g(C_x, C_y) &= \ln S_g - \frac{1}{S_g} \sum_{k=0}^{K-1} \sum_{h=0}^{N-1} |\tilde{g}(k, h; C_x, C_y)|^2 \\
&\quad \cdot \ln |\tilde{g}(k, h; C_x, C_y)|^2 \quad (26)
\end{aligned}$$

where S_g is the image intensity that can be expressed as

$$S_g = \sum_{k=0}^{K-1} \sum_{h=0}^{N-1} |\tilde{g}(k, h; C_x, C_y)|^2. \quad (27)$$

From (25) we can see that, since the phase error compensation term $\exp[-j(C_x h + C_y k)n^2]$ is performed on the phase of $s_p(k; n)$, the signal energy holds constant according to the Parseval theorem [31]. As a result, S_g is independent of (C_x, C_y) . The estimate of (C_x, C_y) is obtained by minimizing the image entropy, expressed as follows:

$$\langle \hat{C}_x, \hat{C}_y \rangle = \arg \min_{C_x, C_y} E_g(C_x, C_y). \quad (28)$$

So far, the optimization based on entropy minimization for estimating the 2-D spatial-variant phase error is established, and it is an optimization function with two parameters. Due to the random characteristics of noise, it contributes a little to the variance of entropy during the phase error estimation. The main contribution would be made by the dominant scatters, as the energy accumulates when the optimal solution achieved, enhancing the sharpness of the image significantly. Therefore, correcting the phase error by minimizing the entropy of the image is reasonable. To solve this optimization, many standard algorithms are available, such as particle swarm optimization and genetic algorithms. However, these nonparametric methods always need great computation time. In this paper, in order to increase the computation efficiency, a gradient-based iterative method is adopted. First, a formula has been derived for the partial derivative of C_x

$$\frac{\partial E_g(C_x, C_y)}{\partial C_x} = -\frac{1}{S_g} \sum_{k=0}^{K-1} \sum_{h=0}^{N-1} [1 + \ln |\tilde{g}(k, h; C_x, C_y)|^2] \cdot \frac{\partial |\tilde{g}(k, h; C_x, C_y)|^2}{\partial C_x} \quad (29)$$

where $|\tilde{g}(k, h; C_x, C_y)|^2 = \tilde{g}(k, h; C_x, C_y) \cdot \tilde{g}^*(k, h; C_x, C_y)$; then, we have

$$\begin{aligned} \frac{\partial |\tilde{g}(k, h; C_x, C_y)|^2}{\partial C_x} &= \tilde{g}^*(k, h; C_x, C_y) \cdot \frac{\partial \tilde{g}(k, h; C_x, C_y)}{\partial C_x} \\ &\quad + \tilde{g}(k, h; C_x, C_y) \cdot \frac{\partial \tilde{g}^*(k, h; C_x, C_y)}{\partial C_x} \\ &= 2 \operatorname{Re} \left[\tilde{g}^*(k, h; C_x, C_y) \cdot \frac{\partial \tilde{g}(k, h; C_x, C_y)}{\partial C_x} \right]. \end{aligned} \quad (30)$$

According to (25), we have

$$\begin{aligned} \frac{\partial \tilde{g}(k, h; C_x, C_y)}{\partial C_x} &= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot s_p(k; n) \\ &\quad \cdot \exp[-j(C_x h + C_y k)n^2] \cdot (-jhn^2). \end{aligned} \quad (31)$$

Similar to the derivation of C_x , the partial derivative of C_y can be obtained. From (29) and (30) we can see that the difference between the derivation of C_x and C_y is the $((\partial \tilde{g}(k, h; C_x, C_y))/\partial)$ term; as to C_y , the expression of this term is expressed as follows:

$$\begin{aligned} \frac{\partial \tilde{g}(k, h; C_x, C_y)}{\partial C_y} &= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot s_p(k; n) \\ &\quad \cdot \exp[-j(C_x h + C_y k)n^2] \cdot (-jkn^2). \end{aligned} \quad (32)$$

With the partial derivative expressions (31) and (32), the gradient of image entropy with respect to (C_x, C_y) is

$$\nabla E_g(C_x, C_y) = \left[\frac{\partial E_g(C_x, C_y)}{\partial C_x}, \frac{\partial E_g(C_x, C_y)}{\partial C_y} \right]^T. \quad (33)$$

In general, the second-order derivative is needed to build the Hessian matrix if the gradient descent algorithms are applied here. However, considering the large computation burden (the

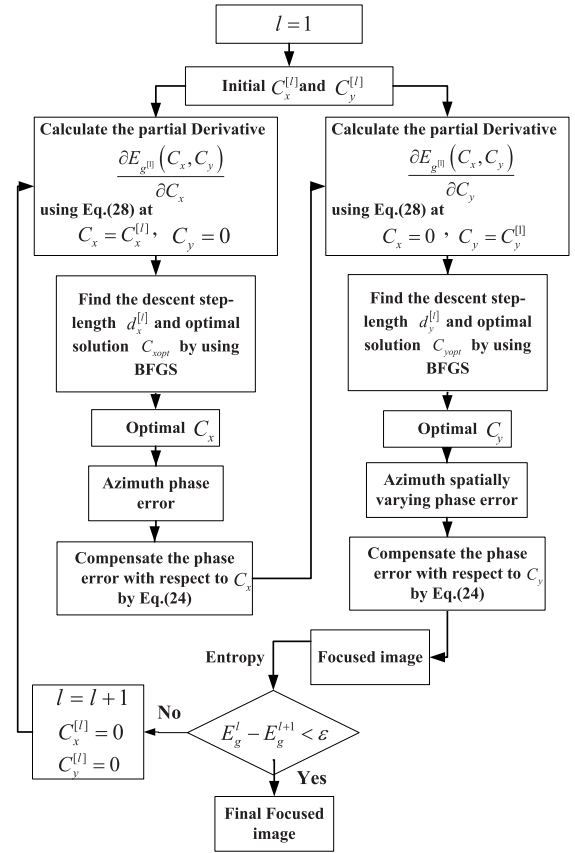


Fig. 2. Flowchart of the coordinate descent algorithm.

second-order derivatives are always complicated), a coordinate descent algorithm implemented by the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm [32], [33] is applied in this paper to search the optimal solution. Herein, we minimized the cost function with respect to a single parameter while holding the other parameter constant to avoid the local optimum. On the other hand, only the first-order derivatives are needed to form the Hessian matrix in the BFGS algorithms, which would reduce the computation burden greatly. The flowchart of the coordinate descent algorithm for searching the optimal solution of the two parameters is presented in Fig. 2.

In general, we usually need some iterations for these two parameters. As a result, we need to perform the phase compensation and image processing several times, and high computational load may be involved. Like any optimization algorithm, a significant aspect on the computational efficiency is the initialization of the parameters (C_x, C_y) . The optimization can be promoted by choosing the initial values of (C_x, C_y) near the optimal solution. What is more, in the low SNR circumstances, a good initialization is essentially significant to avoid trapping into local optimal solution in an optimization-based problem. In Section III-B, a coarse motion estimation for initialization is proposed.

B. Coarse Motion Estimation for Initialization

As discussed earlier, the estimation of the rotational motion parameters (C_x, C_y) can be seen as a global optimization

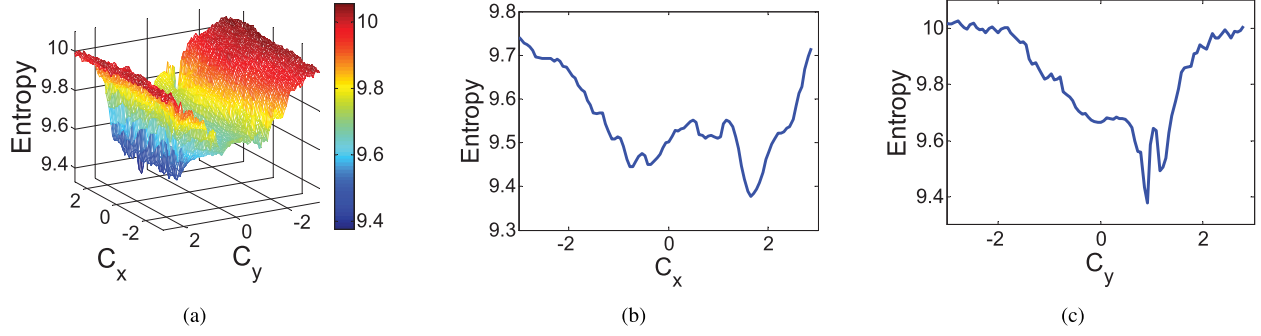


Fig. 3. Variation of entropy with respect to (C_x, C_y) when SNR is -5 dB. (a) Joint variation of entropy with respect to the two parameters. (b) and (c) Independent variation of entropy with respect to one single parameter while holding the other one constant.

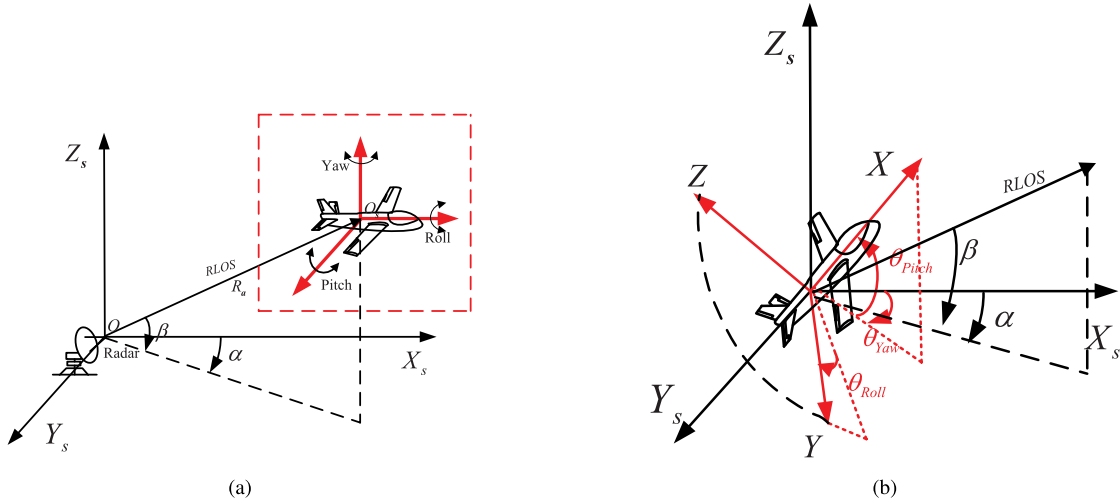


Fig. 4. Coordinate of the 3-D rotation target. (a) Coordinated flight model with roll, pitch, and yaw rotation motion. (b) Euler angles of the commoving frames X_s , Y_s , and Z_s with respect to the fixed radar frames X , Y , and Z .

problem. However, due to the interference of strong noises, the entropy in (26) with respect to a single parameter is usually nonconvex along with extensive local minimums. The variation of entropy with respect to the two parameters (C_x, C_y) is shown in Fig. 3. C_x and C_y are set between -3 and 3 . Fig. 3(b) and (c) shows the independent variation in the entropy with respect to a single parameter while fixing the other one. It is clearly noted that there exist extensive local optimal solutions on both the joint and independent variations. Once the solution is convergent to a local optimal point, it is quite difficult to skip out. Hence, a novel coarse motion estimation method is proposed.

According to Appendix B, $C_x = -((2\pi\zeta_x)/T_a)$ and $C_y = -((2\pi\zeta_y)/\lambda B)$; meanwhile, according to Appendix A, ζ_x and ζ_y are determined by ω_r , ω_p , ω_y , $\dot{\omega}_r$, $\dot{\omega}_p$, $\dot{\omega}_y$, and $\mathbf{i}_{los}$. If these parameters are known already, C_x and C_y can be obtained by using (46) and (49).

Fortunately, Wang *et al.* [34] proposed a method for estimating the orientation of an aerial target based on the velocity measurements by developing an EKF framework, which is a convenient way for us to estimate a coarse candidate of C_x and C_y at a small computational price. As shown in Fig. 4(a), yaw is defined to be the rotation about the vector

(i.e., z-axis in radar frame) coming out the top of the target. It is measured relative to the radar reference coordinate system (X, Y, Z) and modeled as a function of the target motion in the XY plane with the scope of $0^\circ - 360^\circ$. Yaw is expressed by

$$\theta_y = \arctan\left(\frac{v_y}{v_x}\right) \quad (34)$$

where v_x and v_y are the velocities along the x- and y-axes, respectively. A yaw of 0° corresponds to the target flying along the positive direction of X , i.e., $v_y = 0$ in (34).

The target's pitch describes the angle between the total velocity vector and the velocity vector in the XY plane, i.e., θ_{Pitch} shown in Fig. 4(a), at which the altitude of the target changes. Under the coordinated flight model, the pitch can be approximated as

$$\theta_p = \arctan\left(\frac{v_z}{\sqrt{v_x^2 + v_y^2}}\right). \quad (35)$$

Roll is the rotation about the vector extending through the target's nose, as shown in Fig. 4(a), the roll θ_{Roll} is the rotation

about X_s , and it can be approximated by

$$\theta_r = \arctan \left(\frac{|v|^2 \cos(\theta_p)}{Rg} \right) \quad (36)$$

where v is the total velocity of the target and g is the gravity at earth's surface; R is the radius of the curvature of the target's turn and computed as

$$R(t) = \frac{[v_x(t)^2 + v_y(t)^2]^{3/2}}{v_x(t)a_y(t) - a_x(t)v_y(t)} \quad (37)$$

where v_x and v_y are the velocities along the x- and y-axes, and a_x and a_y are the acceleration along the x- and y-axes.

Considering two distinct types of errors (measurement error and modeling error) presenting in the estimation of target's orientation, Wang *et al.* introduced an EKF to balance both types of errors and output the target's orientation accurately. The state model and the measurement model of the EKF are described by

$$\begin{cases} \mathbf{x}_t = \mathbf{x}_{t-1} + \mathbf{q}_t \\ \mathbf{z}_t = \mathbf{h}_t(\mathbf{x}_t) + \mathbf{s}_t \end{cases} \quad (38)$$

where $\mathbf{q}_t$ is the process noise term, modeled as a zero-mean Gaussian random variable with covariance $\mathbf{Q}_t$, $\mathbf{x}_t$ is a vector consisting of the target's yaw, pitch, and roll at time t , $\mathbf{z}_t$ is the measurement vector consisting of target's velocities along the x-, y-, and z-axes, $\mathbf{h}_t(\mathbf{x}_t)$ is a nonlinear mapping of the state into the measurement space, and $\mathbf{s}_t$ is the measurement noise term, modeled as a zero-mean Gaussian random variable with covariance $\mathbf{S}_t$. More details about the filter can be found in [34].

When $\theta_r(t_m)$, $\theta_p(t_m)$, and $\theta_y(t_m)$ are obtained, the RV and RA of the three rotational angles can be achieved by

$$\begin{aligned} \omega &= \frac{d\theta(t_m)}{dt} \\ \dot{\omega} &= \frac{d^2\theta(t_m)}{dt^2}. \end{aligned} \quad (39)$$

Notice that, in Fig. 4(a), the LOS in radar coordinate can be expressed as $\mathbf{i}_{\text{radar}} = [\cos \beta \cos \alpha, \cos \beta \sin \alpha, \sin \beta]^T$, and it can be transformed into target body coordinate [35] as follows:

$$\mathbf{i}_{\text{los}} = \mathbf{rot}[-\theta_r(t_0), -\theta_p(t_0), -\theta_y(t_0)] \cdot \mathbf{i}_{\text{radar}} \quad (40)$$

where $\mathbf{rot}$ denotes the rotational matrix given by (7) and $\theta_r(t_0)$, $\theta_y(t_0)$, and $\theta_p(t_0)$ represent the values of the three rotation angles at initial time t_0 of imaging interval. Using (39) and (40), we can obtain ω_r , ω_p , ω_y , $\dot{\omega}_r$, $\dot{\omega}_p$, $\dot{\omega}_y$, and $\mathbf{i}_{\text{los}}$; then, $C_x = -((2\pi \zeta_x)/T_a)$ and $C_y = -((2\pi c \zeta_y)/\lambda B)$ can be estimated by using (49). It should be pointed out that, although the errors may exist between the orientation obtained by EKF and the true values, the EKF can provide a coarse value of C_x and C_y , which locates near the real ones, and noticeable reduction of the computation time can be realized. What is more, the input of the EKF is the velocities, which are provided by the tracking data, and the output of the filter may be rarely influenced by the low SNR of the received signal. Therefore, the proposed coarse motion estimation is

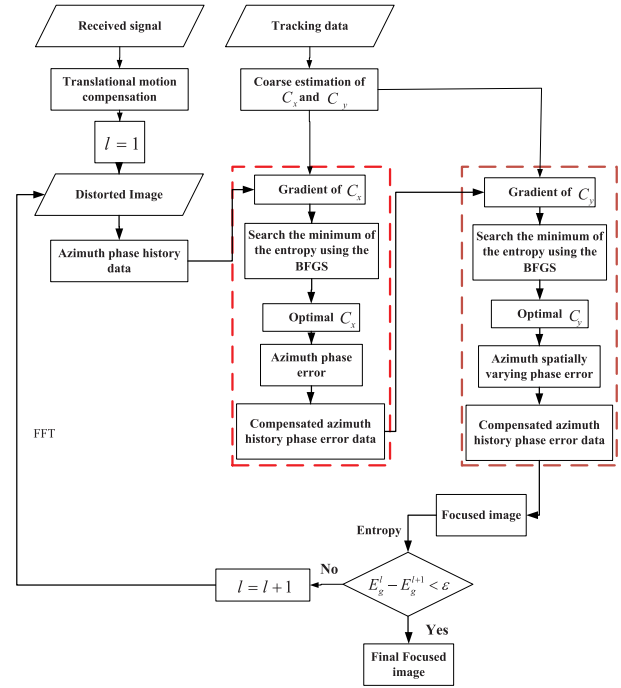


Fig. 5. Flowchart of the proposed method.

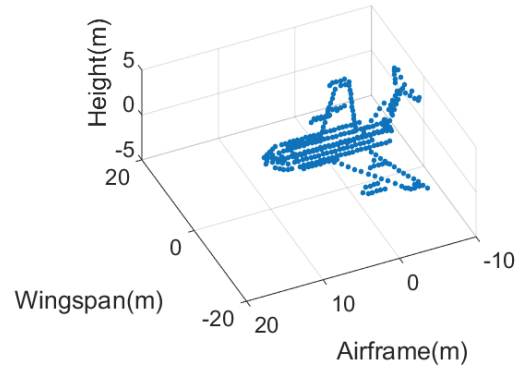


Fig. 6. Scatters distribution of the simulated plane.

suitable in the low SNR situation. For clarity, we present the flowchart of the coordinate descent phase error correction in Fig. 5. From Fig. 5, we note that translational motion compensation is first applied to the received signal. Then, C_x and C_y are estimated by the proposed coarse rotation motion estimation method. And the distorted image is transformed into the azimuth phase history data domain by the fast Fourier transform (FFT), the estimated C_x and C_y values are treated as the initial values, and the optimal C_x and C_y values are estimated iteratively. In every iteration, the phase errors are corrected by the phase error compensation terms respect to C_x and C_y , respectively. Then, a new image is generated, and it becomes the distort image as the input in the next loop, and a new iteration begins. The iteration is running until a convergence is achieved.

IV. PERFORMANCE ANALYSIS

In this section, the performance of the proposed algorithm will be evaluated by using both the simulated data and the real

TABLE I
SOME PARAMETERS OF THE SIMULATED RADAR SYSTEM

Center frequency	5.52 GHz	PRF	400 Hz	Observation time	2.56 s
Bandwidth	400 MHz	Sampling frequency	10 MHz	Pulse width	10 μ s

TABLE II
ESTIMATED MOTION PARAMETERS WHEN TARGET EXECUTES 1-D ROTATIONAL MOTION

Estimated rotational motion parameters					
ω_r	ω_y	ω_p	$\dot{\omega}_r$	$\dot{\omega}_y$	$\dot{\omega}_p$
$-6.6 \times 10^{-4} \text{ rad/s}$	-0.0521 rad/s	0 rad/s	$6.3 \times 10^{-5} \text{ rad/s}^2$	0.0055 rad/s^2	0 rad/s^2
Cx and Cy					
Estimated C_x and C_y			Ideal C_x and C_y		
$C_x=0.2609$	$C_y=-0.1176$		$C_x=0.3440$	$C_y=-0.0725$	

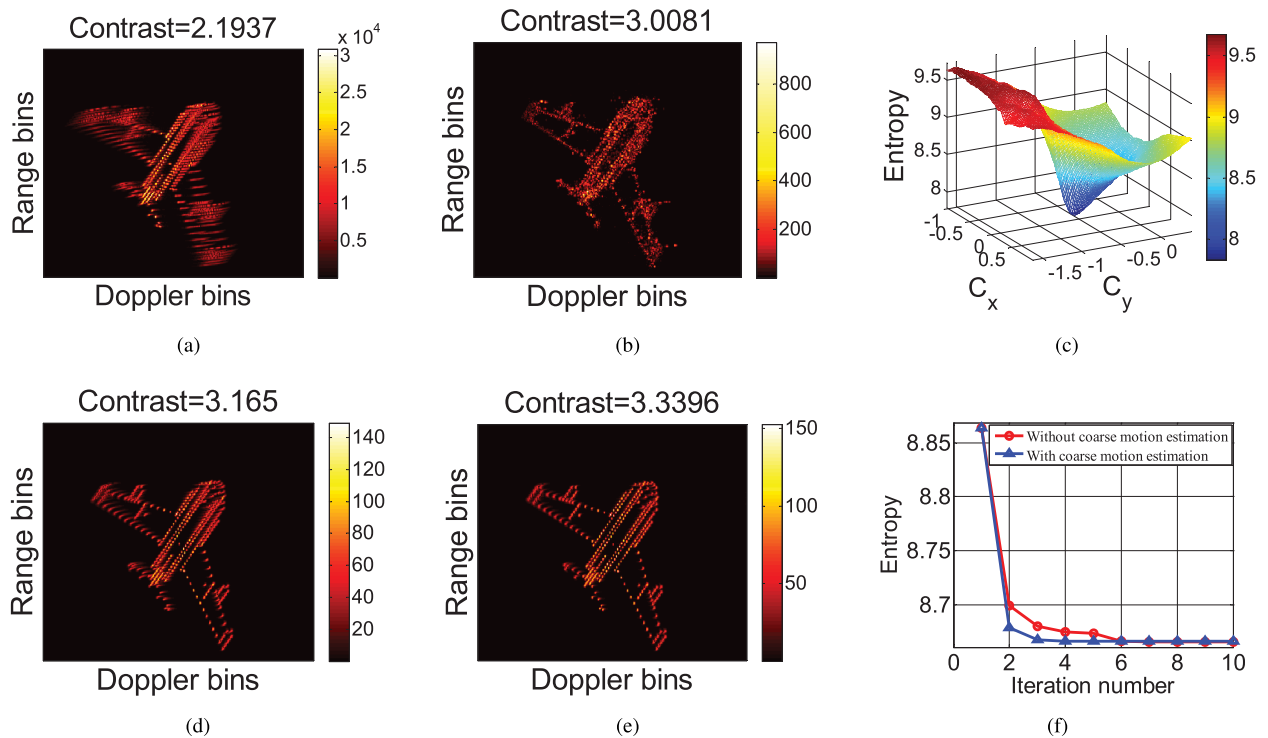


Fig. 7. Experimental results when the target executes 1-D rotational motion. (a) Image obtained by PGA. (b) Image obtained by DechirpClean. (c) Entropy against C_x and C_y . (d) Image obtained by coarse motion compensation. (e) Image obtained by the proposed method. (f) Entropy against iteration number.

measured data. Entropy and contrast of the focused RD images are used as the evaluation criterion of the image quality. The simulated plane data are utilized to analyze the performance of the proposed method. Uniformly accelerated rotation motion and different noise are added into the simulated data, and the phase error compensation is performed by the phase gradient autofocus (PGA) algorithm [36], DechirpClean [11], the coarse motion compensation proposed in Section III-B, and the proposed algorithm. Comparisons are also provided to show the improvement of our method by using the real measured Yak-42 data. All results demonstrate that the enhanced quality is achieved by the proposed method.

TABLE III
COMPUTATION TIME COMPARISON OF INDIVIDUAL METHODS

Algorithms	PGA	DechirpClean	Proposed Method
Computation time [s]	3.19	383.73	9.65

A. Simulation Results

First, some experimental results based on the simulated data are given in the following contents. The scattering model of a plane is distributed in Fig. 6, and some radar system parameters are listed in Table I. The proposed method is applied when the target executes 1-D rotational motion and

TABLE IV
ESTIMATED MOTION PARAMETERS WHEN THE TARGET EXECUTES 3-D ROTATIONAL MOTION

Estimated rotational motion parameters					
ω_r	ω_y	ω_p	$\dot{\omega}_r$	$\dot{\omega}_y$	$\dot{\omega}_p$
$-2.19 \times 10^{-4} \text{rad/s}$	-0.0252rad/s	0.0420rad/s	$2.14 \times 10^{-4} \text{rad/s}^2$	0.0022rad/s^2	-0.0025rad/s^2
Cx and Cy					
Estimated C_x and C_y			Ideal C_x and C_y		
$C_x=0.1811$	$C_y=-0.0742$		$C_x=0.2472$	$C_y=-0.1296$	

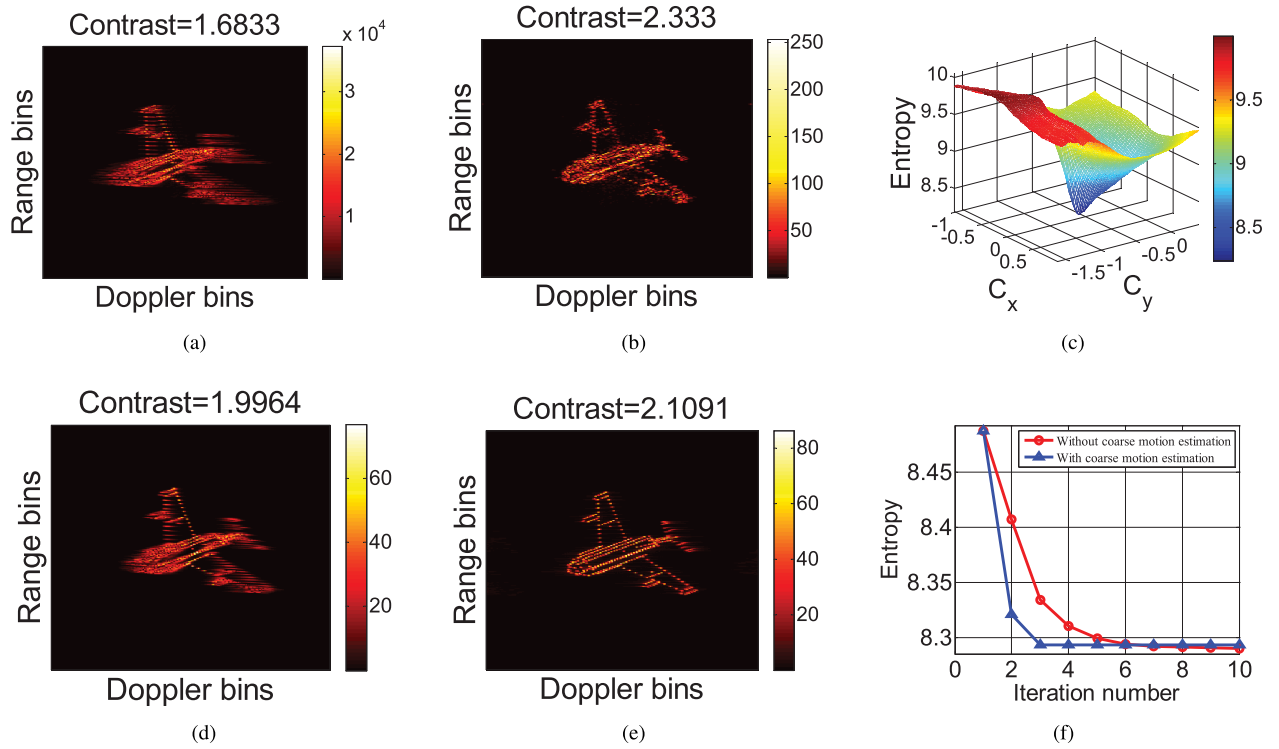


Fig. 8. Experimental results when the target executes 3-D rotational motion. (a) Image obtained by PGA. (b) Image OBTAINED by DechirpClean. (c) Entropy against C_x and C_y . (d) Image obtained by coarse motion compensation. (e) Image obtained by the proposed method. (f) Entropy against iteration number.

3-D rotational motion, respectively. And, for comparison, PGA, DechirpClean, and the coarse compensation proposed in Section III-B are implemented in both the situations.

1) *1-D Rotational Motion:* As to the 1-D rotational motion, the target's velocities along the x-, y-, and z-axes in the simulation are $v_x = 40 + 5 \sin(2\pi t)$, $v_y = 50$, and $v_z = 0$, respectively; 512 pulses are produced during the whole observation time. Fig. 7(c) shows the entropy value with respect to C_x and C_y . It is clearly shown that the entropy value varies smoothly with C_x and C_y and there is no local minimum around the global minimum due to the absence of noise, and the global minimum can be easily obtained by the proposed modified gradient-based parameters estimation method. Using the proposed coarse motion estimation method, Table II gives the coarsely estimated rotational parameters and the ideal ones. As one can note, the estimated C_x and C_y values are much closer to the ideal ones. To show the influence of the coarse motion estimation on the iteration, the entropy against iteration number in the coordinate descent

algorithm is shown in Fig. 7(f). As one can note, when the coarse motion estimation is applied, the iteration reaches the convergence much faster. And thanks to the BFGS algorithms, the CPU time for 10 iterations is 9.38 s as the second-order derivative is not needed. The computation times of each method are compared in Table III in which the running time of the proposed method includes the time of coarse estimation (0.27 s) and the iteration time (9.38 s), and the CPU time is obtained with MATLAB coding using a personal computer with an Intel Core i5 3.30-GHz processor and 8-GB memory. We can observe in Table III that the proposed method yields the significantly enhanced computation efficiency compared with the DechirpClean method.

The images obtained by PGA, DechirpClean, coarse motion compensation, and the proposed method are shown in Fig. 6 (a), (b), (d), and (e). Fig. 7(a) shows that, although the PGA algorithm is applied, due to the presence of range and azimuth spatial-variant phase errors caused by the accelerated rotational motion, the scattering centers on the

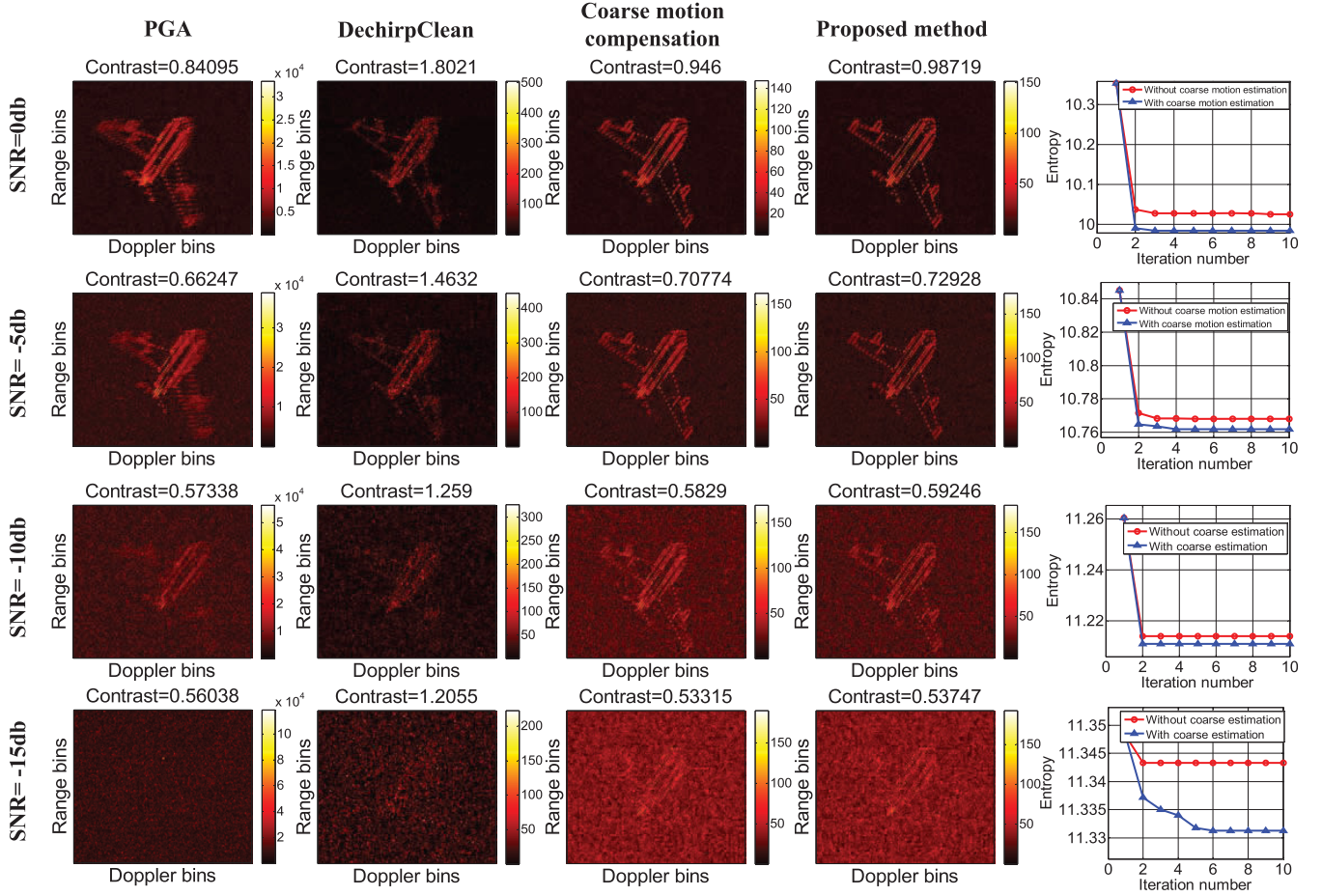


Fig. 9. Experimental results under different SNRs.

wings and nose of the plane are still seriously blurred. The contrast of the image is $\text{Contrast} = 2.1937$. Then, the DechirpClean algorithm is applied. There is a significant improvement in the quality of the image as shown in Fig. 7(b), but the image is still a little blurred. Utilizing the coarse motion estimation and compensating the phase errors with the estimated RV and RA, the compensated image is shown in Fig. 7(d). As we can see, the contrast gets higher as the sharpness of the focused imagery is better than the distorted imagery in Fig. 7(a) and (b). However, the scatterers on the wings and the nose are a little blurred because the 2-D spatial-variant phase errors still exist. The result of the proposed method is shown in Fig. 7(e). The focus accuracy is evidently much better than the imagery using PGA and coarse motion compensation, and the contrast gets much higher ($\text{Contrast} = 3.3396$). The scatterers on the wings and nose are all well focused as the 2-D spatial-variant phase errors compensation is applied to every pixel in the ISAR imagery.

2) *3-D Rotational Motion*: In order to evaluate the performance of the proposed method when the target executes complex 3-D rotational motion, the simulated plane is set to do the 3-D rotational motion with yaw, pitch, and roll. The target's velocities along the x-, y-, and z-axes in the simulation are $v_x = 40 + 5 \sin(2\pi t)$, $v_y = 50 + 5 \cos(2\pi t)$,

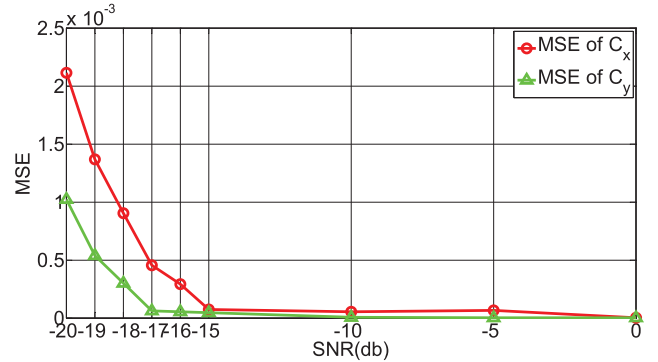


Fig. 10. MSE under different SNRs.

and $v_z = 2 + 4 \sin(2\pi t)$, respectively. The experiment results are shown in Fig. 8. Compared with the experiment results when the target executes 1-D rotational motion, similar conclusions can be drawn from Fig. 8. It is worth mentioning that DechirpClean obtains the ISAR image by applying the FFT to the signal constructed with the estimates of the amplitude and center frequency of all scatterers, and the signal of weak scatterers and the noise will be removed, which leads to the image obtained by DechirpClean a relatively high contrast. The image obtained by DechirpClean is still a little blurred

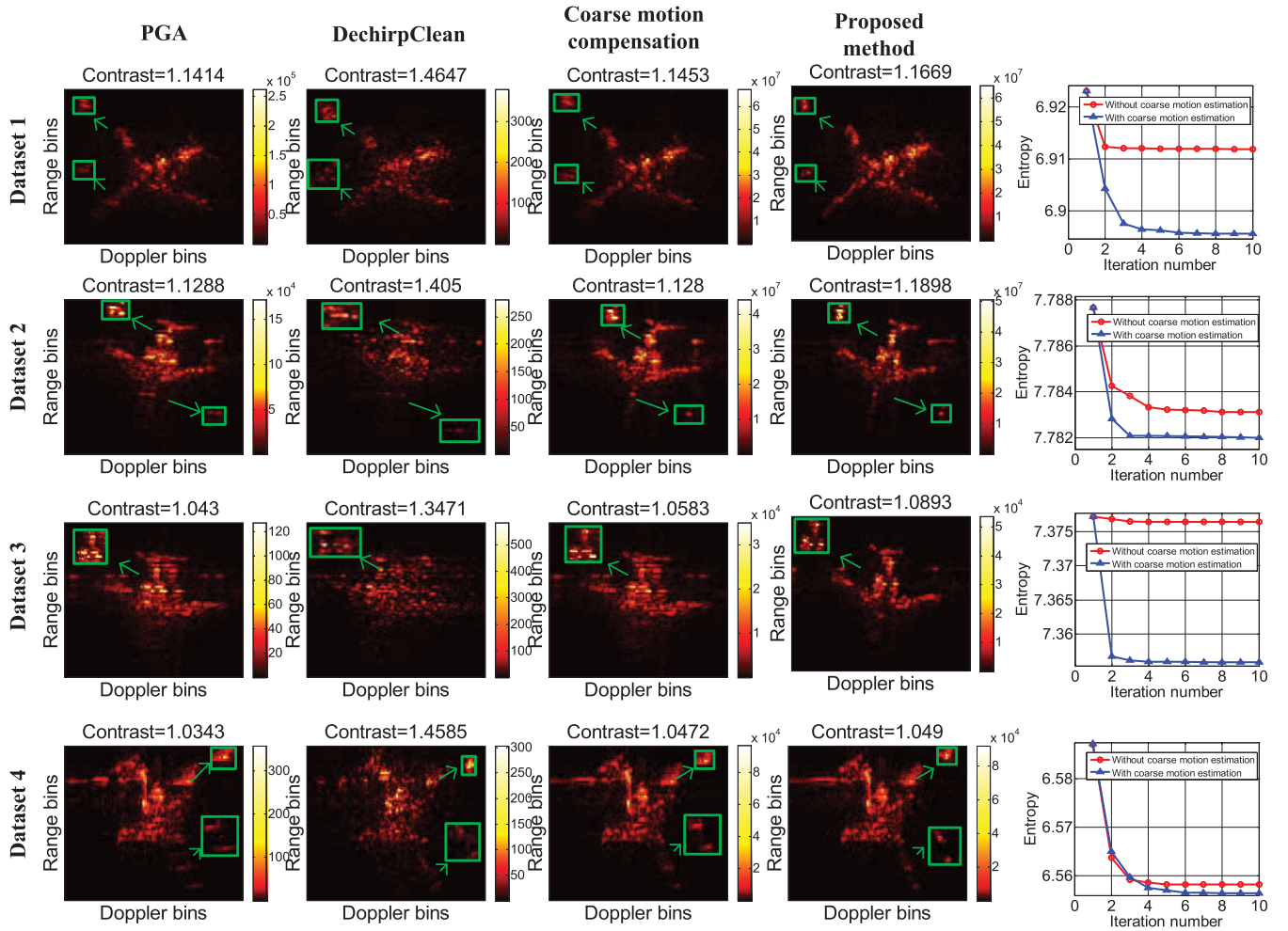


Fig. 11. Experimental results with Yak-42 data set.

compared with the image achieved by the proposed method. Table IV gives the coarsely estimated rotational parameters and the ideal ones, the same as the 1-D rotational motion situation, and the estimated motion parameters via using the proposed coarse motion estimation method are close to the ideal ones. According to the experiment results when the target executes 1-D rotational motion and 3-D rotational motion, the novel phase error signal model proves effective even in the complex 3-D rotational motion situation. And the results of the proposed method also confirm the effectiveness of the modified gradient-based parameters estimation method, as the optimization usually achieves convergence within a few iterations and with low computation burden.

B. Performance Under Different SNRs

In the following contents, the complex white Gaussian noise is added to the simulated signal in the 1-D rotational motion situation to generate different SNRs (from 0 to -15 dB), and the performances of PGA, DechirpClean, coarse motion compensation, and the proposed method will be discussed.

Fig. 9 shows the experiment results under different SNRs. The images compensated by the PGA algorithm are shown in Fig. 9 (left) corresponding to SNR equivalent to 0, -5 , -10 , and -15 dB, respectively. The images obtained

by DechirpClean and coarse motion compensation are shown in the second and third column of Fig. 9 corresponding to different SNRs. The images from the proposed method are given in the fourth column of Fig. 9. And entropy against iteration number in the modified gradient-based parameters estimation method is shown in Fig. 9 (right). As one can note, even the different noises are added, the modified gradient-based optimization usually achieves convergence within less than five iterations, and although the iteration numbers are preset as 10, the CPU times are all about a few seconds. What is more, as we can see, with the assistance of coarse motion estimation, the optimization can reach a better optimum, even the global optimum. In Fig. 9, the images obtained by DechirpClean have high contrasts due to the algorithm's ability of removing certain noise. Although it has high contrasts, the images are still blurred compared with the images obtained by the proposed method. It is notable in Fig. 9 that due to the presence of strong noise, the images obtained by PGA and DechirpClean are poor in quality. The target even cannot be recognized when the SNR declines to -15 dB. On the contrary, the proposed method can generate better images under low SNR, and the clear images can be achieved even when the SNR is -15 dB.

In the experiments, we found that when the SNR decreases below -15 dB, the proposed method will fail to achieve the

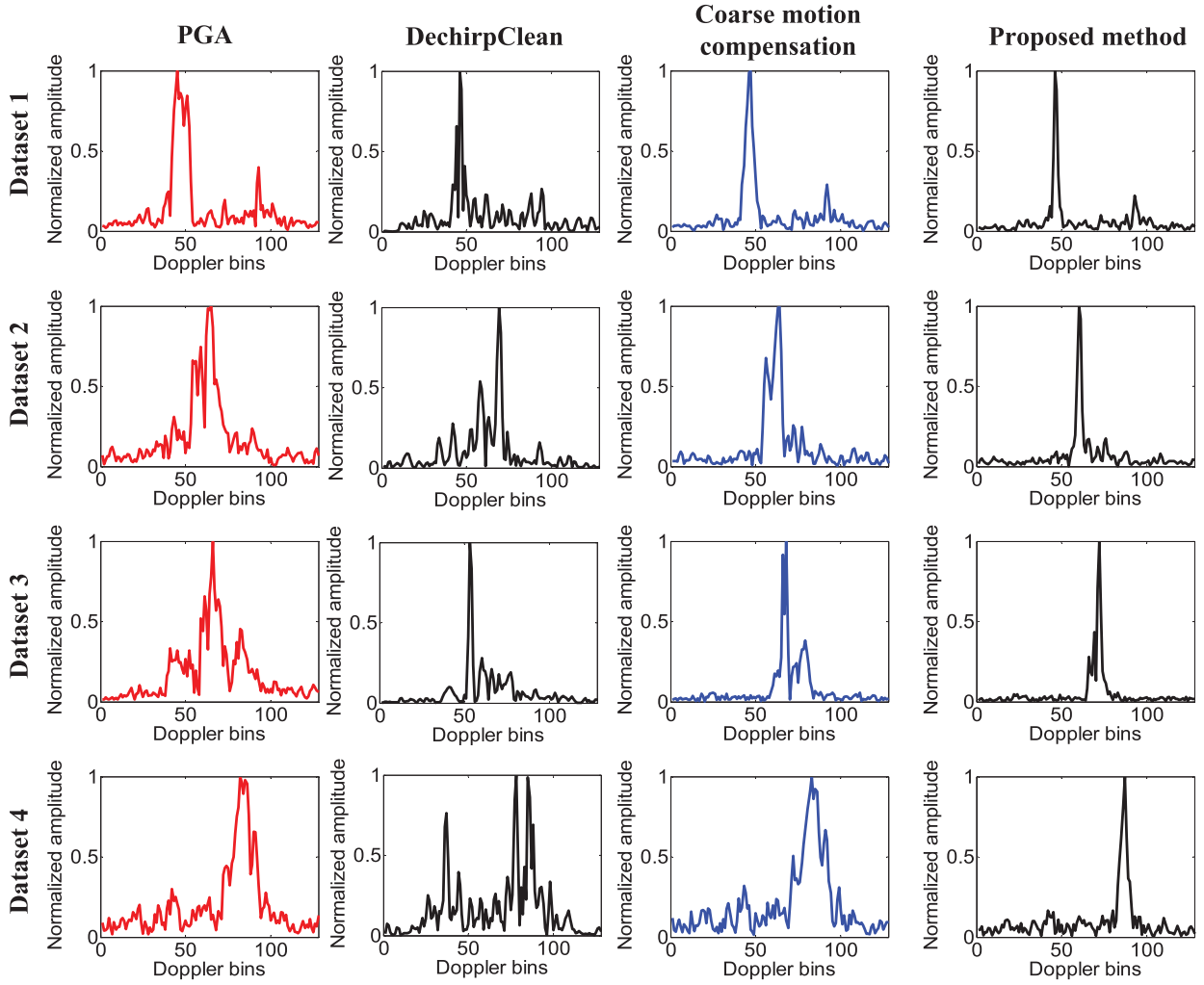


Fig. 12. Images of different range bins.

optimum solution of the two parameters, and the compensated images will be seriously blurred. To show this, the parameters (C_x, C_y) estimated when there is no noise are defined as the ideal ones, and the mean square error between the estimated and ideal parameters is defined by

$$\begin{cases} \text{MSE}_{C_x} = [C_x(k) - C_{x\text{ideal}}]^2 \\ \text{MSE}_{C_y} = [C_y(k) - C_{y\text{ideal}}]^2 \end{cases} \quad (41)$$

where MSE_{C_x} and MSE_{C_y} are the mean square errors of C_x and C_y , respectively. $C_{x\text{ideal}}$ and $C_{y\text{ideal}}$ are the ideal parameters and k represents different SNRs. The MSE corresponding to different SNRs is plotted in Fig. 10. As one can note, the proposed method provides very small MSE only when SNR is above -15 dB. When the SNR decreases below -15 dB, the MSE of the two parameters become much larger, which leads to blurred images. As we mentioned before, the relationship between the focusing quality and image entropy is inconsistent when a very strong noise is involved in the data. And the entropy of image almost relies on the strong noise only, which is independent of the phase error compensation. In this situation, one can use more pulses to obtain high-SNR gain, and then, the well-focused images may be generated by the proposed method.

C. Experiments Using Measured Data

In the following contents, the Yak-42 data are utilized to analyze the performance of the proposed method. In order to illuminate the practicability of the proposed method under different situations, four data sets are selected within which the airplane executes complex rotational motion with different poses. The data sets are recorded by the C-band ISAR experimental system, and the parameters of the radar are the same as that of the simulated radar, which are shown in Table I, except that the pulse repetition frequency is 100 Hz and each data set contains 256 pulses. The zoomed-in view of the image results is shown in Fig. 11.

In Fig. 11, the images obtained by using different data sets are shown in different rows, and the images compensated by different algorithms are shown in different columns; the entropy against iterations number is shown in Fig. 11(right). The contrast is also used to evaluate the focused images in Fig. 11. As we can see clearly, compared with using PGA, DechirpClean, and coarse motion compensation, much more clear images can be achieved by using the proposed method no matter which data set is used. On the contrary, some of the images obtained by DechirpClean are poor in focusing quality although they have high contrasts. To better show

the enhancement of the images, some areas with obvious variation are marked with green boxes in Fig. 11. As one can note, although there is some improvement when coarse motion compensation is applied compared with PGA and DechirpClean, the domain scatters in the green boxes are still blurred and smeared. However, when the proposed method is applied, the blurred domain scatters in the green boxes are well-focused, and although the domain scatters distribute on the different places of the airplane body, the phase errors can be compensated effectively using the proposed phase error model. Fig. 11 (right) shows that the entropy against iterations, and the CPU time for 10 iterations is 8.49, 7.98, 8.01, and 8.81 s respect to the four data sets; similar to the results of simulation, the proposed method can achieve convergence within a few iterations and few seconds. In Fig. 12, to show the improvement on the point spread function by applying the proposed method, we plot the range bins within the green boxes corresponding to different data sets. From Fig. 12, we note that, after the compensation by PGA, DechirpClean, and coarse motion compensation, the responses of the domain scatters still have wide main lobe and negligible sidelobes, which is the reason why the images get blurred and smeared. However, when compensated by our proposed method, the width of the main lobe gets much smaller, and the sidelobes almost all disappear, leaving a single narrow main lobe.

V. CONCLUSION

In this paper, we focus on the range and azimuth spatial-variant phase errors compensation in ISAR imaging of a maneuvering target. Utilizing a novel phase error signal model, the proposed method can compensate the phase errors more accurately. Considering that minimum entropy-based phase error compensation is very robust, an iterative modified gradient-based method is adopted to solve the minimum entropy optimization problem. What is more, to increase the robustness of the proposed method, a novel coarse motion estimation method is proposed to promote the iteration, and the proposed algorithm can reach the global convergence much faster. The experimental results indicate that the imagery entropy of the proposed algorithm approaches a finite positive constant after a few iterations, and a well-focused ISAR image of the maneuvering target will be obtained by the simple but effective algorithm.

APPENDIX A

In (20), $K_0 = i_1 x_p + i_2 y_p$ and $K_1 = (i_2 \omega_y - i_3 \omega_p) x_p + (i_3 \omega_r - i_1 \omega_y) y_p$ can be expressed as

$$\begin{bmatrix} K_0 \\ K_1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x_p \\ y_p \end{bmatrix} \quad (42)$$

where

$$\mathbf{A} = \begin{bmatrix} i_1 & i_2 \\ i_2 \omega_y - i_3 \omega_p & i_3 \omega_r - i_1 \omega_y \end{bmatrix}. \quad (43)$$

Then, K_2 can be expressed as

$$K_2 = \mathbf{q} \begin{bmatrix} x_p \\ y_p \end{bmatrix} \quad (44)$$

where $\mathbf{q}$ is expressed as (50), as shown at the bottom of this page. Using (42) ~ (44), K_2 can be obtained by

$$K_2 = \mathbf{q} \mathbf{A}^{-1} \begin{bmatrix} K_0 \\ K_1 \end{bmatrix}. \quad (45)$$

Then

$$\begin{bmatrix} \zeta_y \\ \zeta_x \end{bmatrix} = [\mathbf{q} \mathbf{A}^{-1}]^T. \quad (46)$$

When matrix $\mathbf{A}$ is irreversible, the quadratic phase error term only related to range bins, which turns the 2-D spatial-variant phase error into 1-D spatial-variant phase error. To compensate the 1-D spatial-variant, readers can refer to [37].

APPENDIX B

Notice that, in (18), after Fourier transform and inverse Fourier transform with respect to f_r and t_m , respectively, K_0 and K_1 reflect the position of the pixel in the image domain. The ideal image can be expressed as

$$\begin{aligned} \tilde{g}(k; h) &= IFFT_{t_m} \{ FFT_{f_r} [\tilde{s}_p(f_r; t_m)] \} \\ &= \sigma_p \text{sinc} \left[\frac{2B}{c} (k - K_0) \right] \cdot \text{sinc} \left[\frac{2T_a}{\lambda} (h - K_1) \right]. \end{aligned} \quad (47)$$

According to the range resolution and cross-range resolution, K_0 and K_1 can be expressed as

$$\begin{aligned} K_0 &= \frac{c}{2B} \cdot k \\ K_1 &= \frac{\lambda}{2T_a} \cdot h \end{aligned} \quad (48)$$

where (k, h) is the pixel position in the image domain, B is the bandwidth, λ is the wavelength, and T_a is the coherent accumulation time. Substituting (48) into (21), $g(k; h)$ can be re-expressed as

$$\begin{aligned} g(k; h) &= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot s_p(k; n) \\ &= \frac{1}{N} \sum_{n=0}^{N-1} \exp \left(j 2\pi \frac{kn}{N} \right) \cdot \tilde{s}_p(k; n) \\ &\quad \cdot \exp[j(C_x h + C_y k)n^2] + \zeta(k; h) \end{aligned} \quad (49)$$

where, $C_x = -((2\pi\zeta_x)/T_a)$ and $C_y = -((2\pi c\zeta_y)/\lambda B)$.

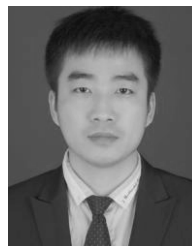
ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their careful review and constructive comments.

$$\mathbf{q} = \left[-i_1 \frac{\omega_p^2 + \omega_y^2}{2} + i_2 \left(\frac{\dot{\omega}_y}{2} + \omega_r \omega_p \right) - i_3 \left(\frac{\dot{\omega}_p}{2} - \omega_r \omega_p \right), -i_1 \frac{\dot{\omega}_y}{2} - i_2 \frac{\omega_r^2 + \omega_y^2}{2} + i_3 \left(\frac{\dot{\omega}_r}{2} + \omega_y \omega_p \right) \right] \quad (50)$$

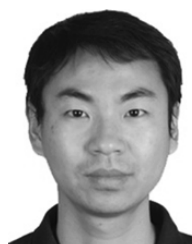
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